USE CASE 1: AI-Assisted Requirements & Architecture Design

Covers:

- Prompting for Developers
- LLM Basics
- AI in Requirements & Design
- ADR-style decisions
- When NOT to use AI

Scenario:

A retail company wants to build a new Online Shopping System with modules:

- Product Catalog
- Cart
- Order Management
- Payment
- User Management

AI is used to:

- Convert business requirements into technical requirements
- Suggest microservice boundaries
- Create ADR documents
- Identify risks

The AI Must:

- Distinguish intent vs instructions
- Suggest architecture (Monolith vs Microservices)
- Recommend database model
- Explain token usage and context length limits
- Provide structured output

Example Prompt (Prompting for Developers)

Bad Prompt (Intent only):

"Design an online shopping system."

Good Prompt (Context + Constraints + Structure):

"You are a senior solution architect.

Design a microservices-based online shopping system.

Constraints:

- Must support 10k concurrent users
- REST APIs only
- Use contract-first design
- Output in structured JSON with: services, responsibilities, API boundaries, risks."

What to Build:

- AI-generated architecture diagram (logical)
- Structured JSON output validation
- ADR document:
    - Why microservices?
    - Why not AI in payment processing logic?
- Requirement traceability document

Deliverables:

- AI-generated architecture JSON
- ADR document
- Requirement breakdown
- Explanation of temperature & token control used